

## GLENANNE NO2, United Kingdom

WMO: 039230

Lat: 54.2369N

Lon: 6.5039W

Elev: 525

StdP: 14.42

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>28.1 | (c)<br>30.3 | (d)<br>23.9                   | (e)<br>18.5 | (f)<br>31.7 | (g)<br>26.1 | (h)<br>20.5 | (i)<br>33.2 | (j)<br>30.7           | (k)<br>46.0 | (l)<br>28.3 | (m)<br>45.6 | (n)<br>6.5               | (o)<br>300 | (p)<br>0.644 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 71.3            | 62.7 | 68.3 | 61.3 | 65.8 | 59.9 | 64.4                | 68.9 | 62.6 | 66.6 | 61.1 | 64.7 | 8.1                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.5                            | 86.3 | 66.4 | 60.9 | 81.5 | 64.5 | 59.5 | 77.3 | 62.9 | 29.7          | 69.0 | 28.4 | 66.6 | 27.4 | 64.7 | 72.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 25.5          | (b) 22.4 | (c) 19.9 | (e) 23.5                   | (f) 76.3 | (g) 4.8            | (h) 3.4 | (i) 20.1                                           | (j) 78.8 | (k) 17.2   | (l) 80.7 | (m) 14.5   | (n) 82.6 | (o) 11.0   | (p) 85.1 |  |  |
|                   |          |          | DB                         | WB       | DB                 | WB      | DB                                                 | WB       | DB         | WB       | DB         | WB       | DB         | WB       |  |  |
|                   |          |          | 23.3                       | 67.5     | 5.0                | 1.9     | 19.7                                               | 68.8     | 16.7       | 69.9     | 13.9       | 71.0     | 10.3       | 72.3     |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.5   | 40.1  | 40.5  | 42.5  | 46.0  | 50.6  | 55.5  | 58.4  | 58.0  | 54.7  | 49.7  | 44.2  | 40.8 |
|                                                                                  | DBStd       | 8.00   | 4.77  | 4.62  | 4.66  | 4.44  | 4.71  | 4.25  | 3.75  | 3.40  | 3.60  | 4.17  | 4.78  | 5.44 |
|                                                                                  | HDD50       | 1523   | 308   | 267   | 234   | 134   | 49    | 4     | 0     | 0     | 5     | 56    | 180   | 286  |
|                                                                                  | HDD65       | 6042   | 773   | 686   | 696   | 570   | 446   | 286   | 210   | 220   | 310   | 473   | 623   | 749  |
|                                                                                  | CDD50       | 964    | 0     | 1     | 3     | 14    | 68    | 169   | 259   | 248   | 145   | 48    | 7     | 2    |
|                                                                                  | CDD65       | 7      | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 37     | 0     | 0     | 0     | 0     | 0     | 11    | 17    | 9     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 1      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 9.5    | 11.5  | 11.4  | 10.5  | 9.6   | 8.6   | 8.1   | 7.6   | 7.8   | 8.4   | 9.7   | 10.4  | 11.0 |
| Precipitation                                                                    | PrecAvg     | 32.2   | 3.1   | 2.2   | 2.6   | 2.3   | 2.4   | 2.4   | 2.1   | 3.0   | 2.8   | 3.3   | 2.9   | 3.1  |
|                                                                                  | PrecMax     | 41.7   | 5.1   | 6.1   | 4.3   | 5.2   | 5.1   | 4.8   | 4.2   | 5.7   | 6.5   | 6.2   | 5.5   | 6.5  |
|                                                                                  | PrecMin     | 23.1   | 1.2   | 0.2   | 1.1   | 0.4   | 0.3   | 0.6   | 0.6   | 0.6   | 0.2   | 1.5   | 1.0   | 0.7  |
|                                                                                  | PrecStd     | 3.9    | 1.0   | 1.3   | 0.9   | 1.1   | 1.2   | 0.9   | 1.0   | 1.2   | 1.4   | 1.5   | 1.2   | 1.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.2  | 53.9  | 57.5  | 63.1  | 70.3  | 74.3  | 76.6  | 74.3  | 68.3  | 61.4  | 57.4  | 55.0 |
|                                                                                  |             | MCWB   | 51.3  | 51.2  | 50.9  | 54.7  | 61.0  | 62.8  | 66.5  | 65.6  | 63.0  | 58.3  | 55.9  | 53.6 |
|                                                                                  | 2%          | DB     | 51.1  | 51.6  | 53.9  | 58.8  | 65.8  | 69.6  | 71.8  | 69.1  | 65.0  | 59.3  | 55.1  | 52.5 |
|                                                                                  |             | MCWB   | 49.5  | 49.6  | 49.9  | 52.3  | 58.2  | 61.1  | 63.3  | 62.6  | 60.5  | 57.0  | 53.8  | 51.2 |
|                                                                                  | 5%          | DB     | 49.4  | 49.8  | 51.8  | 55.7  | 61.9  | 66.2  | 68.3  | 66.4  | 62.8  | 57.7  | 53.1  | 50.8 |
|                                                                                  |             | MCWB   | 48.0  | 47.9  | 48.5  | 50.6  | 55.6  | 59.0  | 61.3  | 61.0  | 58.7  | 55.7  | 51.8  | 49.6 |
|                                                                                  | 10%         | DB     | 47.6  | 47.9  | 49.9  | 53.3  | 58.8  | 63.3  | 65.4  | 64.2  | 60.9  | 56.3  | 51.7  | 48.9 |
|                                                                                  |             | MCWB   | 46.2  | 45.9  | 47.0  | 49.0  | 53.5  | 57.3  | 59.7  | 59.6  | 57.1  | 54.4  | 50.4  | 47.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.5  | 52.1  | 52.9  | 56.3  | 62.6  | 64.9  | 67.2  | 66.6  | 64.0  | 59.4  | 56.1  | 53.8 |
|                                                                                  |             | MCDB   | 52.0  | 53.2  | 55.4  | 60.5  | 68.4  | 71.9  | 74.4  | 71.1  | 66.8  | 60.3  | 56.8  | 54.3 |
|                                                                                  | 2%          | WB     | 49.8  | 50.1  | 50.9  | 53.7  | 59.0  | 62.0  | 64.6  | 63.8  | 61.7  | 57.8  | 54.1  | 51.5 |
|                                                                                  |             | MCDB   | 50.8  | 51.4  | 53.1  | 57.6  | 64.3  | 67.6  | 69.6  | 67.6  | 64.1  | 58.8  | 54.8  | 52.3 |
|                                                                                  | 5%          | WB     | 48.1  | 48.3  | 49.3  | 51.4  | 56.5  | 60.1  | 62.4  | 62.2  | 59.7  | 56.3  | 52.2  | 49.9 |
|                                                                                  |             | MCDB   | 49.1  | 49.7  | 51.3  | 54.7  | 61.0  | 64.9  | 66.8  | 65.2  | 61.9  | 57.5  | 53.1  | 50.9 |
|                                                                                  | 10%         | WB     | 46.3  | 46.3  | 47.4  | 49.6  | 54.4  | 58.3  | 60.7  | 60.6  | 58.0  | 54.7  | 50.5  | 47.8 |
|                                                                                  |             | MCDB   | 47.4  | 47.8  | 49.6  | 52.8  | 57.9  | 62.4  | 64.5  | 63.3  | 60.1  | 56.1  | 51.6  | 48.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.9   | 8.9   | 10.6  | 12.1  | 13.5  | 13.2  | 12.5  | 11.9  | 11.0  | 9.6   | 8.3   | 7.7  |
|                                                                                  |             | MCDBR  | 9.9   | 10.0  | 12.3  | 14.9  | 17.4  | 17.2  | 16.5  | 14.3  | 12.8  | 10.3  | 8.9   | 9.7  |
|                                                                                  |             | MCWBR  | 9.7   | 9.2   | 9.2   | 9.9   | 11.4  | 10.3  | 9.9   | 9.1   | 9.0   | 8.5   | 8.7   | 9.6  |
|                                                                                  | 5% WB       | MCDBR  | 9.8   | 10.0  | 11.4  | 13.9  | 16.2  | 15.7  | 14.9  | 13.1  | 11.9  | 9.8   | 8.7   | 9.7  |
|                                                                                  |             | MCWBR  | 9.8   | 9.6   | 9.4   | 10.1  | 11.2  | 10.4  | 10.2  | 9.5   | 9.4   | 8.9   | 8.8   | 9.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.329  | 0.339 | 0.363 | 0.376 | 0.381 | 0.373 | 0.375 | 0.359 | 0.352 | 0.339 | 0.322 | 0.317 |      |
|                                                                                  | taud        | 2.245  | 2.325 | 2.291 | 2.290 | 2.313 | 2.366 | 2.366 | 2.424 | 2.441 | 2.423 | 2.308 | 2.211 |      |
|                                                                                  | Ebn at Noon | 190    | 236   | 256   | 269   | 274   | 277   | 274   | 273   | 259   | 234   | 193   | 166   |      |
|                                                                                  | Edh at Noon | 20     | 25    | 32    | 36    | 38    | 36    | 36    | 32    | 28    | 23    | 18    | 16    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 197    | 414   | 741   | 1173  | 1482  | 1517  | 1381  | 1136  | 840   | 489   | 245   | 148   |      |
|                                                                                  | RadStd      | 20     | 36    | 77    | 123   | 130   | 158   | 122   | 99    | 42    | 43    | 27    | 16    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (35 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page